TechSolutions Network Infrastructure - Collaboration Department

Student Role: Member 5 - RZ-4 (Sharing/Collaboration)
Project: Systems and Networks Integrated Project 2025/2026
Network: 192.168.42.0/24
Service: NFS Server (Network File System)

PROJECT OVERVIEW

Assigned Network Details (From Summary Table)

Parameter	Value
Student	Yassine Arfaoui
Role	Sharing / Collaboration (RZ-4)
Subnet	/24
Network	192.168.42.0/24
Usable Range	192.168.42.1 — 192.168.42.254
Broadcast	192.168.42.255
Service	NFS Server
Clients	192 PCs (3 for demo)

PHASE 1: GNS3 SETUP AND ROUTER CONFIGURATION

Action 0: Initial Setup

Solution:

Understood role as RZ-4 administrator

Identified network allocation: 192.168.42.0/24

Student	Role	Subnet	Network	Usable range (example)	Broadcast
A	Backbone / Internet admin	/26	192.168.43.0/26	192.168.43.1 — 192.168.43.62	192.168.43.63
B	Web / Marketing (RZ-1)	/19	192.168.0.0/19	192.168.0.1 — 192.168.31.254	192.168.31.255
C	Supervision / IT (RZ-2)	/21	192.168.32.0/21	192.168.32.1 — 192.168.39.254	192.168.39.255
D	Database (RZ-3)	/23	192.168.40.0/23	192.168.40.1 — 192.168.41.254	192.168.41.255
E	Sharing / Collaboration (RZ-4)	/24	192.168.42.0/24	192.168.42.1 — 192.168.42.254	192.168.42.255

Action 1: Install GNS3 and Cisco IOS Image

Steps Completed:

1. **Obtained Cisco IOS image:** c3725-adventerprisek9-mz.124-15.T14.image
2. **Configured router template in GNS3:**
   - Router type: Cisco 3725
   - RAM: 256 MB (reduced from 512 MB due to computer limitations)
   - Platform: c3725
   - Server: Running locally (not GNS3 VM)
   - Idle-PC: 0x60c09aa0 (auto-calculated)

Action 2: Build Network Topology

Devices Added:

1. **RZ-4 Router** (c3725) - 1 unit
2. **Ethernet Switch** (built-in) - 1 unit
3. **VPCS** (Virtual PC Simulator) - 3 units
4. **Cloud Node** (for VM connection) - 1 unit

Connections Made:

From	Interface	To	Interface
RZ-4 Router	Fa0/1	Switch	Ethernet0
Switch	Ethernet1	VPCS1	Ethernet0
Switch	Ethernet2	VPCS2	Ethernet0
Switch	Ethernet3	VPCS3	Ethernet0
Switch	Ethernet4	Cloud1	VMware Network Adapter VMnet1

Note: RZ-4's Fa0/0 left unconnected (reserved for backbone integration with Student A)

Action 3: Configure RZ-4 Router

3.1 Basic Configuration

enable

configure terminal

hostname RZ-4

3.2 Local Network Interface (Fa0/1)

```
interface FastEthernet0/1
 description Connection to Local Switch
 ip address 192.168.42.1 255.255.255.0
 no shutdown
 exit
```

Purpose: Gateway for RZ-4 local network

3.3 DHCP Server Configuration

```
ip dhcp excluded-address 192.168.42.1 192.168.42.50
ip dhcp pool COLLAB_POOL
 network 192.168.42.0 255.255.255.0
 default-router 192.168.42.1
 dns-server 8.8.8.8
 lease 1 0 0
 exit
```

IP Allocation:

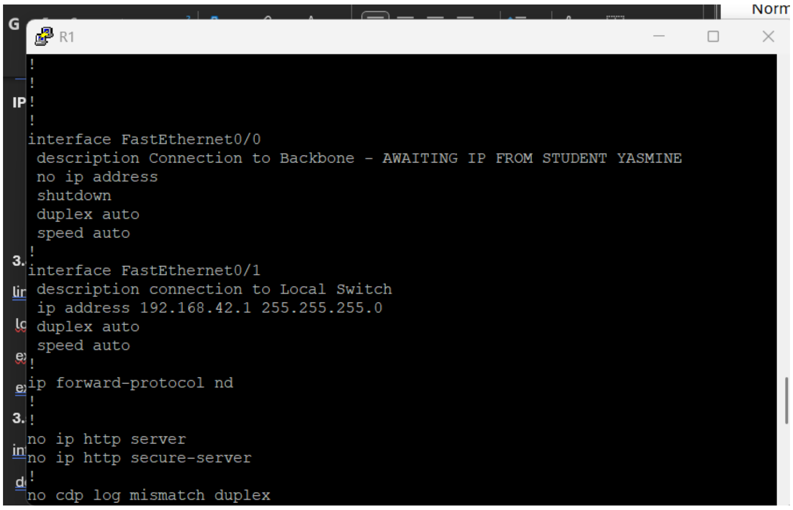
- Reserved (static): 192.168.42.1 - 192.168.42.50
- DHCP Pool: 192.168.42.51 - 192.168.42.254
- Router: 192.168.42.1
- NFS Server: 192.168.42.10 (static, reserved)

3.4 Console Settings

```
line console 0
 logging synchronous
 exec-timeout 0 0
 exit
```

3.5 Backbone Interface (Prepared for Future)

```
interface FastEthernet0/0
 description Connection to Backbone - AWAITING IP FROM STUDENT A
 shutdown
 exit
```



3.6 Save Configuration

end

copy running-config startup-config

Result: Configuration persists across reboots

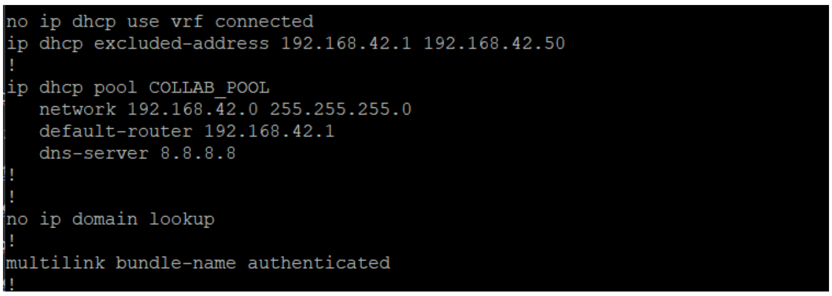
Action 4: Test Client Connectivity

4.1 Configure VPCS Clients

In each VPCS console:

ip dhcp

Results:

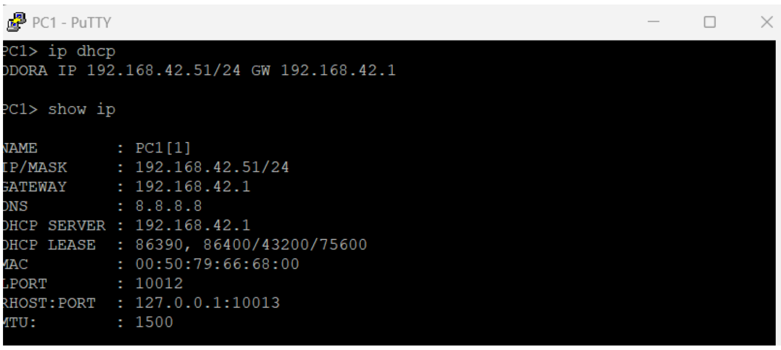
- VPCS1: 192.168.42.51/24
- VPCS2: 192.168.42.52/24
- VPCS3: 192.168.42.53/24
- Gateway: 192.168.42.1
- 

4.2 Connectivity Tests

Test 1: VPCS1 → Router

ping 192.168.42.1

✅ **Result:** Success - 5/5 packets

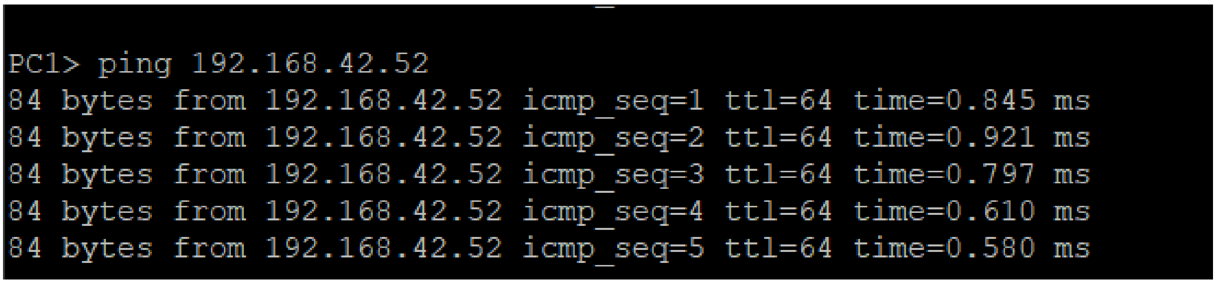


```
PC1 - PuTTY
PC1> ip dhcp
DDORA IP 192.168.42.51/24 GW 192.168.42.1

PC1> show ip

NAME        : PC1[1]
IP/MASK     : 192.168.42.51/24
GATEWAY     : 192.168.42.1
DNS         : 8.8.8.8
DHCP SERVER : 192.168.42.1
DHCP LEASE  : 86390, 86400/43200/75600
MAC         : 00:50:79:66:68:00
LPORT       : 10012
RHOST:PORT  : 127.0.0.1:10013
MTU:        : 1500
```

Test 2: VPCS1 → VPCS2

ping 192.168.42.52

✅ **Result:** Success - inter-client communication works

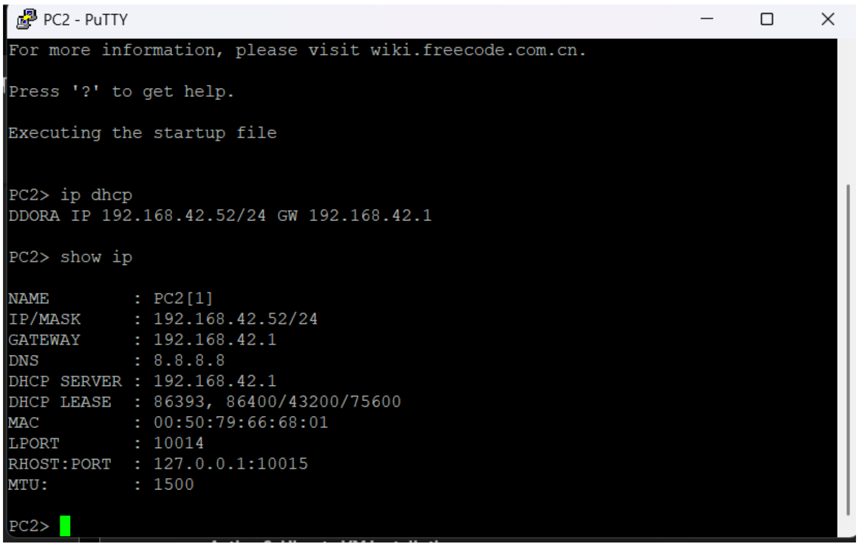


```
PC1> ping 192.168.42.52
84 bytes from 192.168.42.52 icmp_seq=1 ttl=64 time=0.845 ms
84 bytes from 192.168.42.52 icmp_seq=2 ttl=64 time=0.921 ms
84 bytes from 192.168.42.52 icmp_seq=3 ttl=64 time=0.797 ms
84 bytes from 192.168.42.52 icmp_seq=4 ttl=64 time=0.610 ms
84 bytes from 192.168.42.52 icmp_seq=5 ttl=64 time=0.580 ms
```

Test 3: Router → VPCS1

ping 192.168.42.51

✅ **Result:** Success - bidirectional communication confirmed



```
PC2 - PuTTY
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file


PC2> ip dhcp
DDORA IP 192.168.42.52/24 GW 192.168.42.1

PC2> show ip

NAME        : PC2[1]
IP/MASK     : 192.168.42.52/24
GATEWAY     : 192.168.42.1
DNS         : 8.8.8.8
DHCP SERVER : 192.168.42.1
DHCP LEASE  : 86393, 86400/43200/75600
MAC         : 00:50:79:66:68:01
LPORT       : 10014
RHOST:PORT  : 127.0.0.1:10015
MTU:        : 1500

PC2>
```

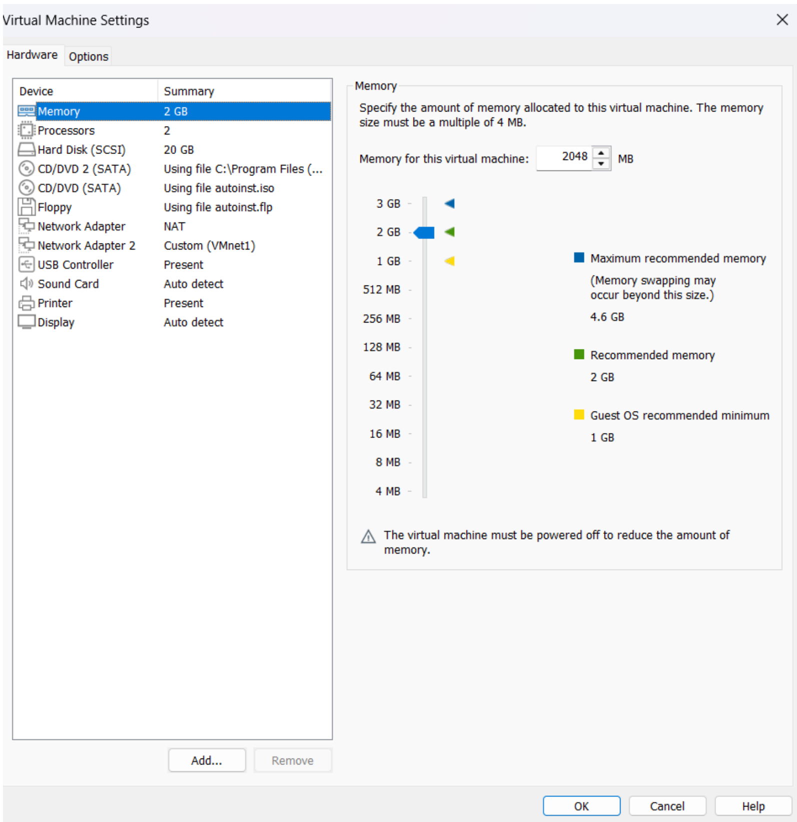
🖥️ PHASE 2: NFS SERVER VM SETUP

Action 5: VMware Network Configuration

Challenge: Ubuntu VM needed to connect to GNS3 topology

5.1 VMware Virtual Network Editor Settings

VMnet1 Configuration:

- **Type:** Host-only
- **Subnet IP:** 192.168.42.0
- **Subnet Mask:** 255.255.255.0
- **DHCP:** Disabled (using RZ-4 router's DHCP)
- **Host virtual adapter:** Enabled
- 

5.2 Windows Host VMnet1 Adapter

Initial Problem: VMnet1 had IP 192.168.42.1 (conflict with router!)

Solution: Changed Windows host VMnet1 adapter to:

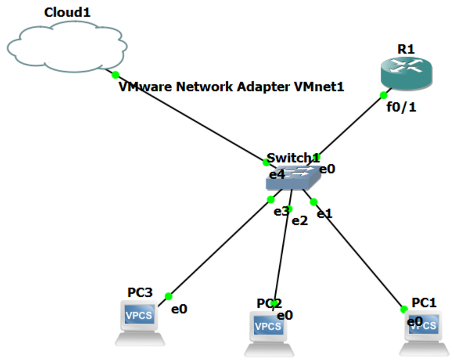
- **IP Address:** 192.168.42.254
- **Subnet Mask:** 255.255.255.0
- **Gateway:** (empty)
- **DNS:** (empty)

Result: IP conflict resolved

5.3 GNS3 Cloud Configuration

Cloud1 Settings:

- **Connected to:** VMware Network Adapter VMnet1 only
- **Purpose:** Bridge between GNS3 and VMware VMs



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Action 6: Ubuntu VM Installation

6.1 VM Specifications

VMware Settings:

- **VM Name:** NFS-Server-RZ4
- **OS:** Ubuntu Server 22.04 LTS
- **RAM:** 1024 MB (1 GB)
- **Disk:** 10 GB (dynamically allocated)
- **CPU:** 1 core

Network Adapters:

1. **Adapter 1 (ens33):** NAT - For internet/updates
2. **Adapter 2 (ens37):** Host-only (VMnet1) - For GNS3 connection

6.2 Ubuntu Installation

Hostname: nfs-server

Packages: Minimal (added NFS later)

Action 7: Configure Static IP on Ubuntu

Challenge: Ubuntu was getting DHCP IP (192.168.42.54) instead of static 192.168.42.10

```
       valid_lft 1063sec preferred_lft 1063sec
    inet6 fe80::20c:29ff:fefb:4184/64 scope link
       valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state
up default qlen 1000
    link/ether 00:0c:29:fb:41:8e brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    inet 192.168.42.10/24 brd 192.168.42.255 scope global noprefixroute e
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fefb:418e/64 scope link
       valid_lft forever preferred_lft forever
```

7.1 Identify Network Interfaces

ip addr show

Found:

- **ens33:** 192.168.33.130/24 (NAT - for internet)
- **ens37:** 192.168.42.54/24 (Host-only - for GNS3) ← Needs static IP

7.2 Configure Netplan

sudo nano /etc/netplan/00-installer-config.yaml

Configuration:

```
network:
 version: 2
 ethernets:
  ens33:
   dhcp4: yes
  ens37:
   dhcp4: no
   addresses:
    - 192.168.42.10/24
   routes:
    - to: default
     via: 192.168.42.1
   nameservers:
    addresses: [8.8.8.8, 8.8.4.4]
```

7.3 Apply Configuration

sudo netplan apply

7.4 Verify Static IP

ip addr show ens37

Result:

- ens37: **192.168.42.10/24** ✅
- Configuration persists after reboot ✅

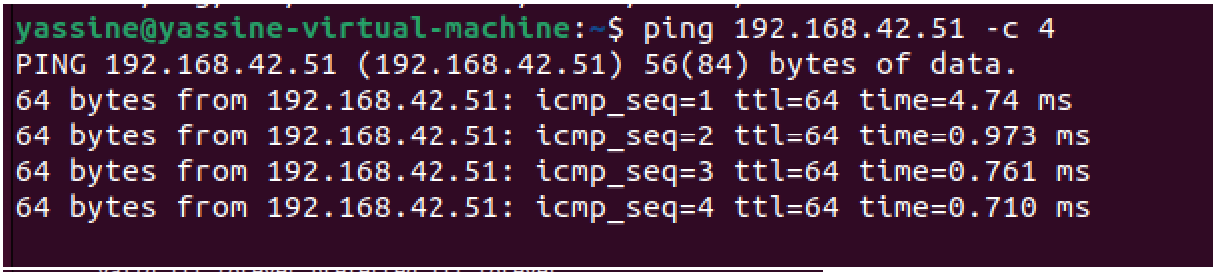
7.5 Test Connectivity

ping 192.168.42.1 -c 4   # Router

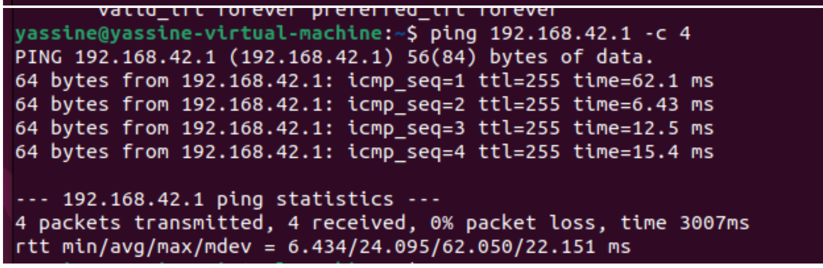
ping 192.168.42.51 -c 4   # VPCS1

ping 8.8.8.8 -c 4   # Internet

All tests successful! ✅



```
yassine@yassine-virtual-machine:~$ ping 192.168.42.51 -c 4
PING 192.168.42.51 (192.168.42.51) 56(84) bytes of data.
64 bytes from 192.168.42.51: icmp_seq=1 ttl=64 time=4.74 ms
64 bytes from 192.168.42.51: icmp_seq=2 ttl=64 time=0.973 ms
64 bytes from 192.168.42.51: icmp_seq=3 ttl=64 time=0.761 ms
64 bytes from 192.168.42.51: icmp_seq=4 ttl=64 time=0.710 ms
```



```
yassine@yassine-virtual-machine:~$ ping 192.168.42.1 -c 4
PING 192.168.42.1 (192.168.42.1) 56(84) bytes of data.
64 bytes from 192.168.42.1: icmp_seq=1 ttl=255 time=62.1 ms
64 bytes from 192.168.42.1: icmp_seq=2 ttl=255 time=6.43 ms
64 bytes from 192.168.42.1: icmp_seq=3 ttl=255 time=12.5 ms
64 bytes from 192.168.42.1: icmp_seq=4 ttl=255 time=15.4 ms

--- 192.168.42.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 6.434/24.095/62.050/22.151 ms
```

🗄️ PHASE 3: NFS SERVER INSTALLATION AND CONFIGURATION

Action 8: Install NFS Server

8.1 Update System

sudo apt update && sudo apt upgrade -y

8.2 Install NFS Kernel Server

sudo apt install nfs-kernel-server -y

Action 9: Configure NFS Share

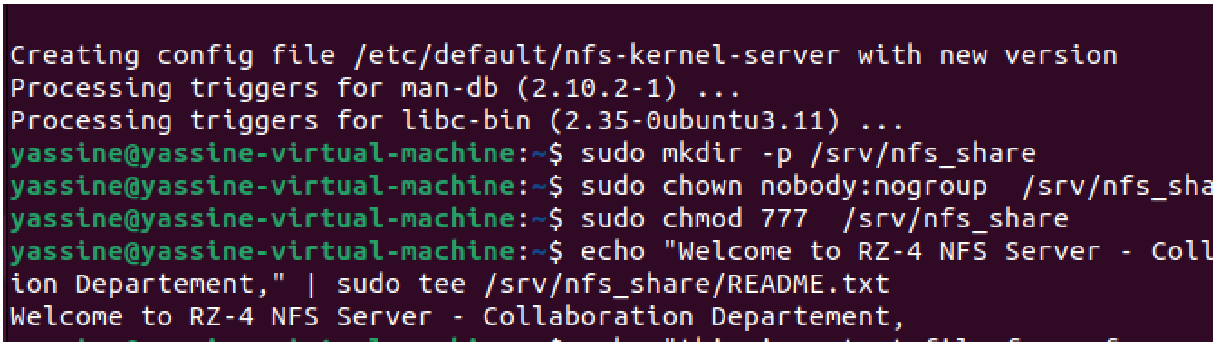
9.1 Create Shared Directory

sudo mkdir -p /srv/nfs_share

9.2 Set Permissions

sudo chown nobody:nogroup /srv/nfs_share sudo chmod 777 /srv/nfs_share

Permissions Explanation:

- **Owner:** nobody (standard for NFS)
- **Group:** nogroup (standard for NFS)
- **Mode:** 777 (read/write/execute for all - testing purposes)
- 

9.3 Create Test Files

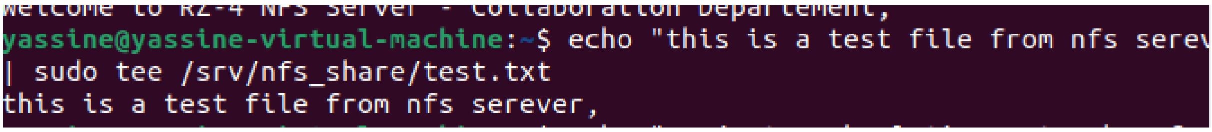
echo "Welcome to RZ-4 NFS Server - Collaboration Department!" | sudo tee /srv/nfs_share/README.txt

echo "This is a test file from NFS server" | sudo tee /srv/nfs_share/test.txt

echo "Project: TechSolutions Network Infrastructure" | sudo tee /srv/nfs_share/info.txt

date | sudo tee /srv/nfs_share/timestamp.txt

Files Created:

- README.txt
- test.txt
- info.txt
- timestamp.txt
- 

Action 10: Configure NFS Exports

10.1 Edit Exports File

sudo nano /etc/exports

Added:

/srv/nfs_share 192.168.42.0/24(rw,sync,no_subtree_check)

Export Options:

- rw - Read and write access
- sync - Synchronous writes (safer, data written immediately)
- no_subtree_check - Performance optimization

10.2 Apply Export Configuration

sudo exportfs -a

10.3 Restart NFS Service

sudo systemctl restart nfs-kernel-server

10.4 Enable NFS at Boot

sudo systemctl enable nfs-kernel-server

Action 11: Verify NFS Server

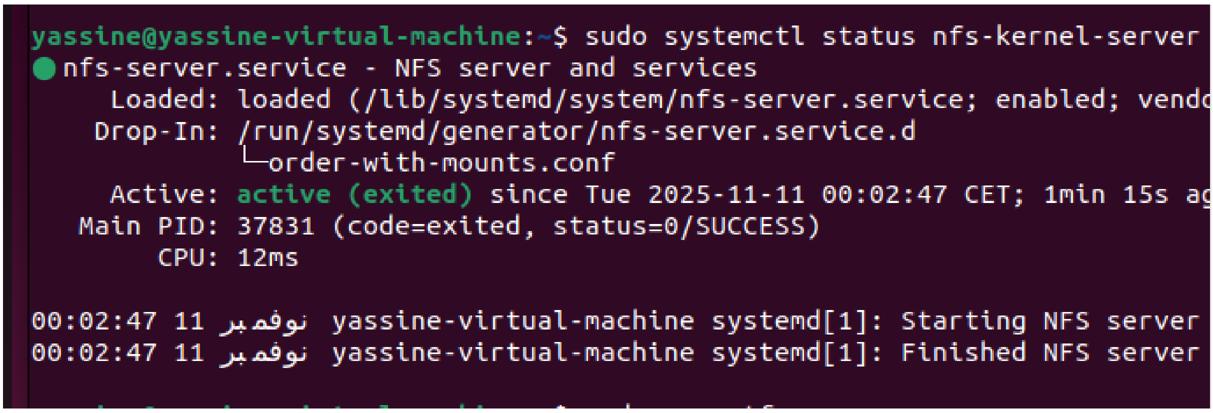
11.1 Check Service Status

sudo systemctl status nfs-kernel-server

Result:

● nfs-server.service - NFS server and services

Active: active (exited) since Tue 2025-11-11 00:02:47 CET



✅ **Service running successfully**

11.2 Verify Exports

sudo exportfs -v

Result:

/srv/nfs_share
192.168.42.0/24(sync,wdelay,hide,no_subtree_check,sec=sys,rw,secure,root_squash,no_all_squash)

✅ **Export configured correctly**

11.3 Check Shared Files

ls -la /srv/nfs_share/

Result:

total 24

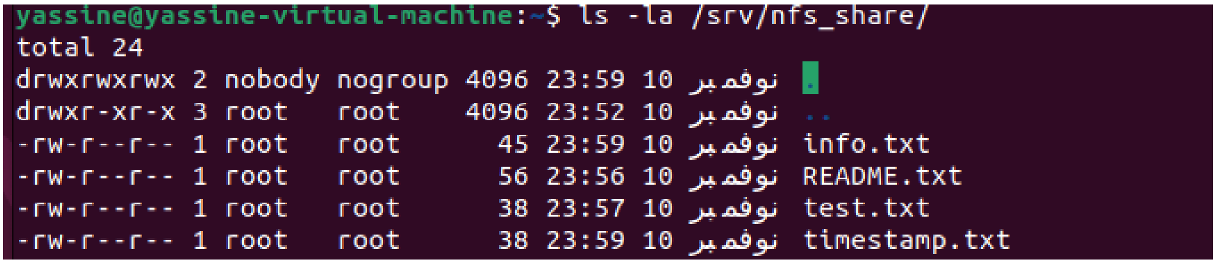
drwxrwxrwx 2 nobody nogroup 4096 ...

-rw-r--r-- 1 root root 56 ... README.txt

-rw-r--r-- 1 root root 38 ... test.txt

-rw-r--r-- 1 root root 45 ... info.txt

-rw-r--r-- 1 root root 38 ... timestamp.txt



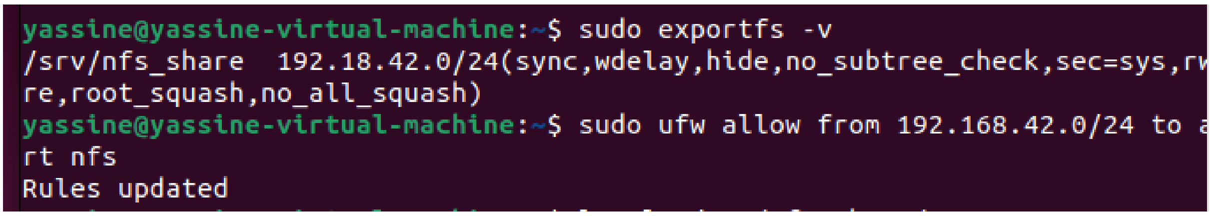
```
yassine@yassine-virtual-machine:~$ ls -la /srv/nfs_share/
total 24
drwxrwxrwx 2 nobody nogroup 4096 23:59 10 نوفمبر .
drwxr-xr-x 3 root   root    4096 23:52 10 نوفمبر ..
-rw-r--r-- 1 root   root      45 23:59 10 نوفمبر info.txt
-rw-r--r-- 1 root   root      56 23:56 10 نوفمبر README.txt
-rw-r--r-- 1 root   root      38 23:57 10 نوفمبر test.txt
-rw-r--r-- 1 root   root      38 23:59 10 نوفمبر timestamp.txt
```

✅ **All files present**

11.4 Configure Firewall

sudo ufw allow from 192.168.42.0/24 to any port nfs

Result: Rules updated ✅ **NFS port accessible from RZ-4 network**



```
yassine@yassine-virtual-machine:~$ sudo exportfs -v
/srv/nfs_share  192.18.42.0/24(sync,wdelay,hide,no_subtree_check,sec=sys,rw
re,root_squash,no_all_squash)
yassine@yassine-virtual-machine:~$ sudo ufw allow from 192.168.42.0/24 to a
rt nfs
Rules updated
```

✅ **PHASE 4: FINAL TESTING AND VERIFICATION**

Action 12: End-to-End Connectivity Tests

Test Matrix

Test	From	To	Command	Result
1	Ubuntu VM	RZ-4 Router	ping 192.168.42.1	✅ 0% loss
2	Ubuntu VM	VPCS1	ping 192.168.42.51	✅ 0% loss
3	Ubuntu VM	Internet	ping 8.8.8.8	✅ 0% loss
4	VPCS1	RZ-4 Router	ping 192.168.42.1	✅ 0% loss
5	VPCS1	NFS Server	ping 192.168.42.10	✅ 0% loss

Test	From	To	Command	Result
6	VPCS2	NFS Server	ping 192.168.42.10	✅ 0% loss
7	VPCS3	NFS Server	ping 192.168.42.10	✅ 0% loss
8	RZ-4 Router	NFS Server	ping 192.168.42.10	✅ 100% success

```
yassine@yassine-virtual-machine:~$ echo "==================================" | sudo tee /srv/nfs_share/SERVER_INFO.txt
echo "RZ-4 NFS Server Information" | sudo tee -a /srv/nfs_share/SERVER_INFO.txt
echo "==================================" | sudo tee -a /srv/nfs_share/SERVER_INFO.txt
echo "" | sudo tee -a /srv/nfs_share/SERVER_INFO.txt
echo "Server IP: 192.168.42.10" | sudo tee -a /srv/nfs_share/SERVER_INFO.txt
echo "Network: 192.168.42.0/24" | sudo tee -a /srv/nfs_share/SERVER_INFO.txt
echo "Gateway: 192.168.42.1 (RZ-4 Router)" | sudo tee -a /srv/nfs_share/SERVER_INFO.txt
echo "Shared Directory: /srv/nfs_share" | sudo tee -a /srv/nfs_share/SERVER_INFO.txt
echo "Access: Read/Write for entire network" | sudo tee -a /srv/nfs_share/SERVER_INFO.txt
echo "Department: Collaboration / Sharing (RZ-4)" | sudo tee -a /srv/nfs_share/SERVER_INFO.txt
echo "" | sudo tee -a /srv/nfs_share/SERVER_INFO.txt
echo "To mount from clients:" | sudo tee -a /srv/nfs_share/SERVER_INFO.txt
echo "sudo mount 192.168.42.10:/srv/nfs_share /mnt/nfs_share" | sudo tee -a /srv/nfs_share/SERVER_INFO.txt
echo "" | sudo tee -a /srv/nfs_share/SERVER_INFO.txt
echo "Created: $(date)" | sudo tee -a /srv/nfs_share/SERVER_INFO.txt
==================================
RZ-4 NFS Server Information
==================================

Server IP: 192.168.42.10
Network: 192.168.42.0/24
Gateway: 192.168.42.1 (RZ-4 Router)
Shared Directory: /srv/nfs_share
Access: Read/Write for entire network
Department: Collaboration / Sharing (RZ-4)

To mount from clients:
```

```
==================================
RZ-4 NFS Server Information
==================================

Server IP: 192.168.42.10
Network: 192.168.42.0/24
Gateway: 192.168.42.1 (RZ-4 Router)
Shared Directory: /srv/nfs_share
Access: Read/Write for entire network
Department: Collaboration / Sharing (RZ-4)

To mount from clients:
sudo mount 192.168.42.10:/srv/nfs_share /mnt/nfs_share
```

R1

```
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    200.100.4.0/24 is directly connected, FastEthernet0/0
O    200.100.5.0/24 [110/84] via 200.100.4.1, 00:03:43, FastEthernet0/0
C    192.168.42.0/24 is directly connected, FastEthernet0/1
     200.100.0.0/30 is subnetted, 10 subnets
O       200.100.0.36 [110/74] via 200.100.4.1, 00:03:43, FastEthernet0/0
O       200.100.0.32 [110/74] via 200.100.4.1, 00:03:43, FastEthernet0/0
O       200.100.0.12 [110/74] via 200.100.4.1, 00:03:43, FastEthernet0/0
O       200.100.0.8 [110/138] via 200.100.4.1, 00:03:45, FastEthernet0/0
O       200.100.0.4 [110/138] via 200.100.4.1, 00:03:45, FastEthernet0/0
O       200.100.0.0 [110/138] via 200.100.4.1, 00:03:45, FastEthernet0/0
O       200.100.0.28 [110/138] via 200.100.4.1, 00:03:45, FastEthernet0/0
O       200.100.0.24 [110/74] via 200.100.4.1, 00:03:45, FastEthernet0/0
O       200.100.0.20 [110/138] via 200.100.4.1, 00:03:45, FastEthernet0/0
O       200.100.0.16 [110/138] via 200.100.4.1, 00:03:50, FastEthernet0/0
O    200.100.1.0/24 [110/84] via 200.100.4.1, 00:03:50, FastEthernet0/0
S    200.100.2.0/24 [1/0] via 200.100.4.1
S    200.100.3.0/24 [1/0] via 200.100.4.1
     10.0.0.0/24 is subnetted, 1 subnets
```

R1

```
O       10.0.0.0 [110/85] via 200.100.4.1, 00:03:50, FastEthernet0/0
O    192.168.32.0/24 [110/84] via 200.100.4.1, 00:03:50, FastEthernet0/0
S    200.100.0.0/16 [1/0] via 200.100.4.254
O    192.168.40.0/23 [110/84] via 200.100.4.1, 00:03:50, FastEthernet0/0
O    192.168.0.0/19 [110/84] via 200.100.4.1, 00:03:50, FastEthernet0/0
O    192.168.32.0/21 [110/94] via 200.100.4.1, 00:03:50, FastEthernet0/0
RZ-4#ping 200.100.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.100.1.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 240/341/420 ms
RZ-4#ping 200.100.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.100.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 80/137/284 ms
RZ-4#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.4.4.4          1   FULL/DR         00:00:35    200.100.4.1     FastEthernet0/
0
RZ-4#
```

R1

```
O       10.0.0.0 [110/85] via 200.100.4.1, 00:03:50, FastEthernet0/0
O    192.168.32.0/24 [110/84] via 200.100.4.1, 00:03:50, FastEthernet0/0
S    200.100.0.0/16 [1/0] via 200.100.4.254
O    192.168.40.0/23 [110/84] via 200.100.4.1, 00:03:50, FastEthernet0/0
O    192.168.0.0/19 [110/84] via 200.100.4.1, 00:03:50, FastEthernet0/0
O    192.168.32.0/21 [110/94] via 200.100.4.1, 00:03:50, FastEthernet0/0
RZ-4#ping 200.100.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.100.1.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 240/341/420 ms
RZ-4#ping 200.100.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.100.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 80/137/284 ms
RZ-4#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.4.4.4          1   FULL/DR         00:00:35    200.100.4.1     FastEthernet0/
0
RZ-4#
```